

Net Detective

Unit 10: Time Capsule Engineers · Lesson 10-3 · EduWonderLab

UNIT 10 · LESSON 10-3 · TEACHER NOTES

Standard 6.G.4

FORMAT Match solids to nets

TIME 25–35 min

GROUPING Pairs

TOPIC Surface Area Using Nets

STANDARD 6.G.4

PREP LEVEL Medium — print net cards and solid cards

Learning goal *I can use a net to find the surface area of a solid.*

Teacher setup

- Print one set per pair. Students match each solid to its net.
- They label matching faces, find each face area, and add for surface area.
- One card has an invalid net to catch.

Materials

Net cards, Solid cards, Colored pencils, Face-area table, Pencil.

Differentiation & language support

- Face-area table lists each face and its area (SPED).
- Sentence stem: “The surface area is the sum of ____ faces.”
- ESOL: color matching faces the same color.

Key vocabulary (English / Español):

Term	Español	What it means
net	plantilla	a flat pattern that folds into a solid
surface area	área superficial	the total area of all faces

Quick teacher check: *Check that students caught the invalid net on Card 5.*

Net Detective

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Name: _____ Date: _____ Class: _____

Your goal: *I can use a net to find the surface area of a solid.*

What you need

Net cards, Solid cards, Face-area table.

How to play

1. Match a solid to its net.
2. Label each face and find its area.
3. Add the face areas for the surface area.
4. Record the total in square units.

Net Cards

A net shows every face. Find each face area and add them for the surface area.

1. A cube with edge 3. Surface area?	2. A box $4 \times 3 \times 2$. Surface area?
3. A box $5 \times 2 \times 2$. Surface area?	4. A cube with edge 5. Surface area?
5. Invalid-net card: a cube net shows only 5 faces. Valid?	6. A box $6 \times 3 \times 1$. Surface area?

Face-area table (each face + total):

Explain your thinking

Explain why a net makes it easier to find surface area.

Sentence frame: *The surface area is the sum of ___ faces.*

★ **Challenge:** *Card 2 and Card 6 both equal a surface area near 52–54. Which is greater, and by how much?*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Standard 6.G.4

Answers

1. A cube with edge 3. Surface area? Answer: 54 sq units (6×9)	2. A box $4 \times 3 \times 2$. Surface area? Answer: 52 sq units
3. A box $5 \times 2 \times 2$. Surface area? Answer: 48 sq units	4. A cube with edge 5. Surface area? Answer: 150 sq units (6×25)
5. Invalid-net card: a cube net shows only 5 faces. Valid? Answer: No — a cube needs 6 faces	6. A box $6 \times 3 \times 1$. Surface area? Answer: 54 sq units

Teaching notes & look-fors

- Surface area of a box = $2(lw + lh + wh)$.
- Card 2: $2(12 + 8 + 6) = 52$; Card 6: $2(18 + 6 + 3) = 54$.